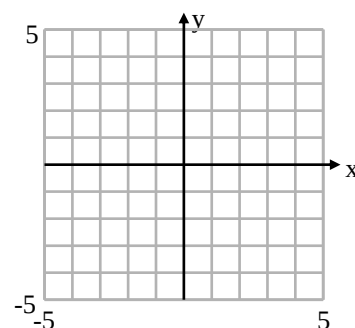


Absolute Value Functions

Preview:

Example 1: Graph $f(x) = \begin{cases} x; & x > 0 \\ 0; & x = 0 \\ -x; & x < 0 \end{cases}$



*The entire graph is on/above the ___-axis which means that all the x-values are _____.

*What mathematical operation always has a positive outcome? _____

** You just graphed the equation of the absolute value function: $f(x) = |x|$

General Equation of an Absolute Value Function

$$y = a|x - h| + k$$

*(h,k) is the vertex (the point of the graph) of the function.

* The line of symmetry is the line $x = h$.

* Always V-shaped

* Opens up if _____

* Opens down if _____

* The graph is wider than the graph of $y = |x|$ if _____

The graph is narrower than the graph of $y = |x|$ if _____ (the v-shape is steeper)

The graph is the same width of $y = |x|$ if _____

Graphing an Absolute Value Function

1. Find and plot the vertex.
2. Sketch the line of symmetry using a dashed line.
3. Plot two points on one side of the line of symmetry.
4. Use the line of symmetry to plot two points on the other side to complete the V-shape.

Example 2: Graph $y = -|x - 1| + 1$

$$y = a|x - h| + k$$

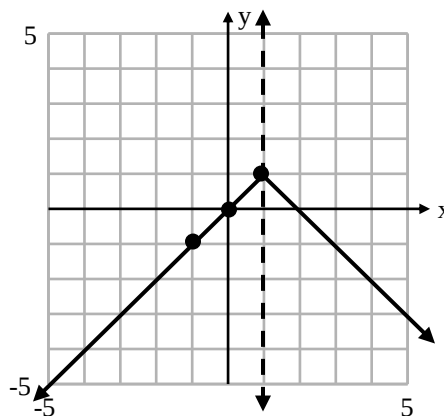
Vertex (h, k) = (1,1)

Line of symmetry ($x = h$); $x = 1$

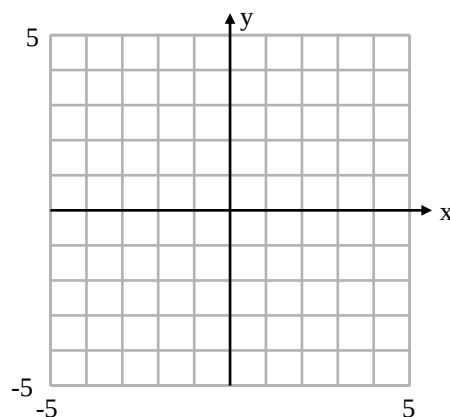
Opens down since $a < 0$ ($-1 < 0$)

Same width of $y = |x|$ since $|a| = |-1| = 1$

x	$y = - x - 1 + 1$	y
0	$y = - 0 - 1 + 1$	0
-1	$y = - -1 - 1 + 1$	-1



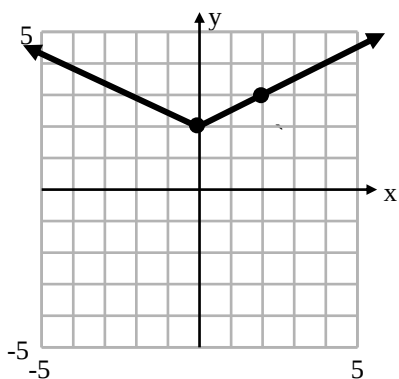
Example 3: Graph $y = |x - 2| - 3$



Writing an Absolute Value Function

1. Identify the vertex
2. Plug the vertex into the equation
3. Identify a point on the graph
4. Plug the point into the equation to find a
5. Plug the vertex and a into the general equation to get your answer.

Example 4: Write the equation shown below.



Vertex: (0,2)

$y = a|x - h| + k$ plug-in vertex

$y = a|x - 0| + 2$ simplify

$y = a|x| + 2$

plug-in point on the graph (2,3)

$3 = a|2| + 2$ solve for a .

$3 = 2a + 2$

$1 = 2a$

$\frac{1}{2} = a$

Plug in a and the vertex into the general equation

$y = a|x - h| + k$

$y = \frac{1}{2}|x - 0| + 2$ simplify

$y = \frac{1}{2}|x| + 2$

Example 5: Write the equation shown below.

